

[--- CODE CELL ---]

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[--- CODE CELL ---]

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

[--- CODE CELL ---]

```
np.random.seed(42)
```

[--- CODE CELL ---]

```
X = 2 * np.random.rand(100,1)
y = 4 + 3 * X + np.random.randn(100,1)
df = pd.DataFrame({ "X": X.flatten(), "y": y.flatten() })
```

[--- CODE CELL ---]

```
df.head()
```

[--- CODE CELL ---]

```
df.describe()
```

[--- CODE CELL ---]

```
sns.scatterplot(x="X", y="y", data = df)
plt.title("Scatter plot of X vs y")
plt.show()
```

[--- CODE CELL ---]

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

[--- CODE CELL ---]

```
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred = lr.predict(X_test)
```

[--- CODE CELL ---]

```
print("Slope (w):", lr.coef_[0][0])
print("Intercept (b):", lr.intercept_[0])
```

[--- CODE CELL ---]

```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("MSE:", mse)
print("R2 Score:", r2)
```

[--- CODE CELL ---]

```
#plt.scatter(X_train, y_train, label = "Actual")
```

```
plt.scatter(X_test, y_test, label = "Actual")
plt.plot(X_test, y_pred, color="red", label = "Prediction")
plt.legend()
plt.title("Sklearn Linear Regression")
plt.show()
```

[--- CODE CELL ---]